

$S(x)$: x is a sophomore $J(x)$: x is a junior $Se(x)$: x is a senior
 $F(x)$: x is a freshman Domain: Set of all students in the discrete maths class.

- (1) $\exists x(M(x) \wedge F(x))$ (2) $\exists x(M(x) \wedge S(x))$ (3) $\exists x(C(x) \wedge S(x))$
 (4) $\exists x(M(x) \wedge J(x))$ (5) $\exists x(C(x) \wedge J(x))$ (6) $\exists x(C(x) \wedge Se(x))$ are all respectively True
 (7) $S(x) \Leftrightarrow \neg F(x) \wedge \neg J(x) \wedge \neg Se(x)$ is True (8) $F(x) \Leftrightarrow \neg S(x) \wedge \neg J(x) \wedge \neg Se(x)$ is True
 (9) $J(x) \Leftrightarrow \neg F(x) \wedge \neg S(x) \wedge \neg Se(x)$ is True (10) $Se(x) \Leftrightarrow \neg F(x) \wedge \neg S(x) \wedge \neg J(x)$ is True
 (11) $M(x) \Leftrightarrow \neg C(x)$ is True

a) There is a student in the class who is a junior: $\exists x J(x)$
 From (4), we know $\exists x(M(x) \wedge J(x))$ is True. By defn. of existential quantifier, there exists a student a in the class s.t. $M(a) \wedge J(a)$ is True. By defn. of AND, $M(a)$ is True and $J(a)$ is True. By defn. of existential quantifier, $\exists x J(x)$ is True

b) Every student in the class is a computer science major.
 $\forall x(C(x))$
 From (1), we know $\exists x(M(x) \wedge F(x))$ is True. By defn. of existential quantifier, there exists a student a in the class s.t. $M(a) \wedge F(a)$ is True. By defn. of AND, $M(a)$ and $F(a)$ are both respectively True.

From (11), we know $\forall x(M(x) \Leftrightarrow \neg C(x))$ is True. By defn. of universal quantifier, for all students b in the class, $M(b) \Leftrightarrow \neg C(b)$ is True. $\therefore M(a) \Leftrightarrow \neg C(a)$ is True. By defn. of biconditional, $\neg C(a)$ is True. By defn. of negation, $\neg(\neg C(a))$ is False. From the double negation law, $C(a)$ is False. From the defn. of universal quantifier, $\forall x(C(x))$ is False.

c) There is a student in the class who is neither a mathematics major nor a junior: $\exists x(\neg M(x) \wedge \neg J(x))$